

UNIVERSITY OF COLORADO BOULDER

College of Engineering and Applied Science
Department of Computer Science

CSCI 3022: Introduction to Data Science

with Probability and Statistics

Summer 2026 Course Calendar

Tuesday, Wednesday & Thursday, 9:45–10:55 AM

Engineering Center ECES 112 — In Person

Course Information

- **Course:** CSCI 3022 — Introduction to Data Science with Probability and Statistics
- **Section:** 400 **Class Number:** 16198
- **Credit Hours:** 3
- **Session:** Boulder D Session — 10 Weeks
- **Course Dates:** May 26, 2026 through July 31, 2026
- **Instruction Mode:** In Person
- **Campus:** Boulder Main Campus
- **Meeting Days/Time:** Tuesday, Wednesday, Thursday — 9:45–10:55 AM
- **Location:** Engineering Center, ECES 112

Important Dates

- **First Day of Class:** Tuesday, May 26, 2026
- **Independence Day:** Saturday, July 4, 2026 (falls on a weekend — no class meeting affected)
- **Exam 1:** Thursday, June 18, 2026 (Week 4)
- **Exam 2:** Thursday, July 9, 2026 (Week 7)
- **Project Part 1 Due:** Friday, July 17, 2026 (Week 8)
- **Project Part 2 Due:** Thursday, July 30, 2026 (Week 10, last class day)

- **Last Day of Classes / Final Exam:** Thursday, July 30, 2026
- **Session Ends:** Friday, July 31, 2026

Course Schedule

Note: This schedule is subject to change. Students will be notified of any updates via Canvas announcements. All assignments are due by 11:59 PM on the specified date unless otherwise noted. Class meets **Tuesday, Wednesday, and Thursday** each week, 9:45–10:55 AM in Engineering Center ECES 112. **There are no Friday classes.**

Independence Day: July 4, 2026 falls on a Saturday. No class meeting is affected (T/W/Th schedule). HW 9 (Week 6) is set to **Thursday, July 2** to keep its deadline off the July 4 weekend.

Wk	Date	Topic/Activity	Assignments
1	Tue, May 26	Course Introduction & Lesson 1 Course overview, syllabus, expectations Lesson 1: Introduction to Data Science First half of Lesson 2	
1	Wed, May 27	Lessons 2 & 3 Finish Lesson 2 All of Lesson 3 (<i>Lesson 4 is an assigned video included in HW 2</i>)	
1	Thu, May 28	Lesson 5 & First Half of Lesson 6 All of Lesson 5 First half of Lesson 6	HW 1 Due
1	Sat, May 30	(<i>No class — assignment deadline only</i>)	HW 2 Due (<i>includes L4 video</i>)
2	Tue, Jun 2	Finish Lesson 6 & Start Lesson 7 Finish Lesson 6 (approx. 25 minutes) First 50 minutes of Lesson 7	
2	Wed, Jun 3	Lesson 7 (cont.) & Lesson 8 Finish Lesson 7 (<i>Simulating Probabilities section is an assigned video in HW 4 — skip in class</i>) First half of Lesson 8	
2	Thu, Jun 4	Lesson 8 & Lesson 10 Finish Lesson 8 All of Lesson 10 (<i>Lesson 9 is an assigned video included in HW 5</i>)	HW 3 Due
2	Sat, Jun 6	(<i>No class — assignment deadline only</i>)	HW 4 Due (<i>includes L7 Simulating Probabilities video</i>)

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Wk	Date	Topic/Activity	Assignments
3	Tue, Jun 9	Lesson 11 & First Half of Lesson 12 All of Lesson 11 First half of Lesson 12	
3	Wed, Jun 10	Lesson 12 (cont.) & Lesson 13 Finish Lesson 12 All of Lesson 13	
3	Thu, Jun 11	Lesson 14 & First Half of Lesson 15 All of Lesson 14 First half of Lesson 15	HW 5 Due <i>(includes L9 video)</i>
3	Sat, Jun 13	<i>(No class — assignment deadline only)</i>	HW 6 Due
4	Tue, Jun 16	Lesson 15 & Lesson 17 Finish Lesson 15 Start Lesson 17 <i>(Lesson 16 is an assigned video included in HW 7)</i>	
4	Wed, Jun 17	Lesson 17 (cont.) & Exam 1 Review Finish Lesson 17 Exam 1 Review Session Q&A and practice problems	
4	Thu, Jun 18	EXAM 1 In-class exam (full class period, 9:45–10:55 AM) Topics: Lessons 1–17 + assigned video lessons (L4, L7 Simulating Probabilities, L9)	
5	Tue, Jun 23	Lesson 18 All of Lesson 18 <i>(Lesson 19 is an assigned video included in HW 8)</i>	
5	Wed, Jun 24	Lesson 20 All of Lesson 20	
5	Thu, Jun 25	Lesson 21 All of Lesson 21 <i>(Skip section connecting hypothesis tests and confidence intervals — covered in HW 9 video)</i>	HW 7 Due <i>(includes L16 video)</i>
5	Sun, Jun 28	<i>(No class — assignment deadline only)</i>	HW 8 Due <i>(includes L19 video)</i>
6	Tue, Jun 30	Lesson 22	

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Wk	Date	Topic/Activity	Assignments
6	Wed, Jul 1	Lesson 23 & Start Lesson 24 All of Lesson 23 Start Lesson 24	
6	Thu, Jul 2	Finish Lesson 24 Finish Lesson 24	HW 9 Due <i>(moved off Sat Jul 4 / holiday weekend) (includes L21 hypothesis-test/CI video)</i>
6	Sat, Jul 4	INDEPENDENCE DAY (HOLIDAY) No class meeting (weekend).	
7	Tue, Jul 7	Lesson 25 (First Half) & Exam 2 Review First half of Lesson 25 Exam 2 Review Session Q&A and practice problems	
7	Wed, Jul 8	Finish Lesson 25 Finish Lesson 25 Pre-Exam 2 Q&A	
7	Thu, Jul 9	EXAM 2 In-class exam (full class period, 9:45–10:55 AM) Topics: Lessons 18–25 + assigned video lessons (L16, L19, L21 hypothesis-test/CI)	
8	Tue, Jul 14	Lesson 26	
8	Wed, Jul 15	Lesson 27	HW 10 Due <i>(includes L25 video portion)</i>
8	Thu, Jul 16	Lesson 28	
8	Fri, Jul 17	<i>(No class — assignment deadline only)</i>	Project Part 1 Due
9	Tue, Jul 21	Lesson 29	
9	Wed, Jul 22	Lesson 30	
9	Thu, Jul 23	Lesson 31	
10	Tue, Jul 28	Lesson 32 Final lesson; comprehensive wrap-up of course material	

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Wk	Date	Topic/Activity	Assignments
10	Wed, Jul 29	Final Exam Review Comprehensive review of all course topics Q&A session	
10	Thu, Jul 30	FINAL EXAM (IN-CLASS) Comprehensive exam covering all course topics In-class during regular class period (9:45–10:55 AM) LAST DAY OF CLASSES	Project Part 2 Due (11:59 PM)
10	Fri, Jul 31	Session Ends	

Assessment Schedule

Homework Assignments

Homework assignments are due by 11:59 PM on the specified date. Deadlines fall on Thursdays (post-class) and Saturdays/Sundays (weekend deadlines) to give students recovery time after class meetings.

- **HW 1:** Due Thursday, May 28, 2026 *Introduction & Lessons 1–3*
- **HW 2:** Due Saturday, May 30, 2026 *Lessons 1–5 + Lesson 4 video*
- **HW 3:** Due Thursday, June 4, 2026 *Lessons 5–7*
- **HW 4:** Due Saturday, June 6, 2026 *Lessons 7–8, 10 + Lesson 7 Simulating Probabilities video*
- **HW 5:** Due Thursday, June 11, 2026 *Lessons 11–13 + Lesson 9 video*
- **HW 6:** Due Saturday, June 13, 2026 *Lessons 13–14*
- **HW 7:** Due Thursday, June 25, 2026 *Lessons 15–18 + Lesson 16 video*
- **HW 8:** Due Sunday, June 28, 2026 *Lessons 18, 20, 21 + Lesson 19 video*
- **HW 9:** Due **Thursday, July 2, 2026** *Lessons 22–24 + Lesson 21 hypothesis-test/CI video (set before the July 4 weekend)*
- **HW 10:** Due Wednesday, July 15, 2026 *Lesson 25 + Lesson 25 video portion (gives a full week after Exam 2)*

Note: Lessons 4, 9, 16, 19, the Simulating Probabilities section of Lesson 7, the hypothesis-tests/CI section of Lesson 21, and a portion of Lesson 25 are delivered as *assigned video lessons* bundled with the corresponding homework. Video lessons are required.

Data Science Project

- **Project Part 1:** Due Friday, July 17, 2026 (11:59 PM) — Week 8
- **Project Part 2:** Due **Thursday, July 30, 2026** (11:59 PM) — Week 10, final class day

The project applies CSCI 3022 methods (data wrangling, probability, statistics, simulation, inference, and basic statistical modeling) to a dataset or problem of your choosing. Detailed prompts, deliverables, and rubrics are posted on Canvas. **Weeks 8–10 are intentionally lighter on homework so you can focus on the project.**

Exams

- **Exam 1: Thursday, June 18, 2026** (In-class, 9:45–10:55 AM)
Topics: Lessons 1–17 plus assigned video lessons covered in HW 1–6 (L4, L7 Simulating Probabilities, L9)
- **Exam 2: Thursday, July 9, 2026** (In-class, 9:45–10:55 AM)
Topics: Lessons 18–25 plus assigned video lessons covered in HW 7–9 (L16, L19, L21 hypothesis-test/CI)
- **Final Exam: Thursday, July 30, 2026** (In-class, 9:45–10:55 AM)
Comprehensive — covers all course topics (Lessons 1–32 and all assigned video lessons)

Course Description

Introduces students to the tools, methods, and theory behind extracting insights from data. Covers algorithms for cleaning and munging data, probability theory and common distributions, statistical simulation, drawing inferences from data, and basic statistical modeling. Same as CSPB 3022.

Prerequisites

- (CSCI 2270 or CSCI 2275) and
- (APPM 1360 or MATH 2300) and
- (CSCI 2824 or ECEN 2703 or APPM 3170 or MATH 2001)

All prerequisite courses require a minimum grade of **C–**. Students may be administratively dropped for non-attendance, non-participation, or lack of prerequisites.

Important Notes

- All homework and project assignments are due at 11:59 PM on the specified date unless otherwise noted on Canvas.
- Several lessons (Lessons 4, 9, 16, 19, and portions of L7, L21, and L25) are delivered as *assigned video lessons* bundled with the corresponding homework. Plan accordingly — video lessons are required.
- Make-up exams will only be allowed for documented emergencies or university-excused absences. Contact the instructor in advance whenever possible.
- This schedule is subject to change. All adjustments will be announced on Canvas.
- Class meets Tuesday, Wednesday, and Thursday only. There are no Friday class meetings.
- Independence Day (July 4, 2026) falls on a Saturday, so no class meeting is affected. HW 9 has been set to Thursday, July 2, 2026 so the deadline doesn't land on the July 4 weekend.
- Project Part 2 is due Thursday, July 30, 2026 (the final class day, same date as the Final Exam — exam is in-class, project submission is via Canvas).
- All course materials, assignments, and announcements will be posted on Canvas.

Last Updated: May 25, 2026

University of Colorado Boulder
Department of Computer Science
Instructor: Dr. Osita Onyejekwe